

Physics for Scientists

Science

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Physics for Scientists (A11385)

Short Title: Physics for Scientists
Faculty Science and Computing
Department Science
Cluster Mathematics and Physics
Author CKEARY
Credits 5

Level: Introductory

Description of Module / Aims

This module introduces the student to the fundamentals of thermal physics, electromagnetism and geometrical optics. The module will also introduce the fundamentals of atomic and nuclear physics. The module contains a practical component, where the student can develop measurement, logical thinking, analysis and report-writing skills.

Programmes

PHYI-0002	BSc (Common Entry) (WD_SSCIE_D)	1 / 2 / M
PHYI-0002	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	1 / 2 / M
PHYI-0002	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	1 / 2 / M
PHYI-0002	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	1 / 2 / M
PHYI-0002	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	1 / 2 / M
PHYI-0002	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	1 / 2 / M
PHYI-0002	BSc (Hons) in Science (Common Entry) (WD_SSCCM_B)	1 / 2 / M
PHYI-0002	BSc in Food Science with Business (WD_SFDSC_D)	1 / 2 / M
PHYI-0002	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	1 / 2 / M
PHYI-0002	BSc in Pharmaceutical Science (WD_SPHSC_D)	1 / 2 / M

Indicative Content

- Thermal physics: temperature and temperature scales; expansion of solids and liquids; heat capacity and change of state; ideal gas laws; thermal transfer
- Electromagnetism: electrostatics; DC components, circuits and circuit laws; magnetism and electromagnetism; thermoelectric effects
- Geometrical optics and properties of light: reflection and refraction, lenses and mirrors, ray tracing, the lens and mirror equation & lens power; refraction, critical angle and total internal reflection
- Atomic physics: photoelectric emission
- Nuclear physics: radioactivity and radioactive decay; nuclear reactions and radiation hazards; fundamental forces

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe and explain the fundamental principles of the physics of heat and temperature.
2. Describe and explain the fundamental principles of electromagnetism.
3. Complete routine calculations on simple electrical circuits.
4. Explain how lenses & mirrors form images and describe the properties of light.
5. Describe and explain the fundamental principles of atomic and nuclear physics.
6. Apply problem-solving skills to generate solutions to problems relating to electromagnetism, optics, thermal, atomic and nuclear physics.
7. Complete experiments, record and analyse experimental data and present results in the form of a report.

Learning and Teaching Methods

- Lectures: The lectures will present new topics and their related concepts. Problem solving techniques will also be used in class to analyse physical situations and find numerical solutions.
- Tutorials: Online tutorials linked to core text.
- Laboratory-based practicals: An integrated practical programme, designed to run in parallel with the lectures, allows the student to develop a range of experimental skills. The importance of planning of experiments, data acquisition and analysis will be emphasised.
- Self-directed study.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	40%	1,2,3,4,5,6
Continuous Assessment	60%	
<i>Practical</i>	40%	7
<i>Tutorial/Problem Sheets</i>	20%	6

Assessment Criteria

- <40%:** Inability to demonstrate a basic understanding of the fundamental principles of thermal physics, electromagnetism, optics, atomic and nuclear physics. Inability to carry out practical skills in the physics laboratory to an adequate level.
- 40%-49%:** Ability to demonstrate a basic understanding of the fundamental concepts of thermal physics, electromagnetism, optics, atomic and nuclear physics. Ability to provide partial solutions to numerical problems. Has a basic level of competency in the required skills in the physics laboratory.
- 50%-59%:** Ability to clearly describe and define concepts encountered in thermal physics, electromagnetism, optics, atomic and nuclear physics. Ability to analyse physical situations and apply suitable problem-solving techniques to find numerical solutions. Good practical skills in the physics laboratory, good clear reporting of practical work.
- 60%-69%:** In addition to the above, ability to apply problem-solving techniques to new similar problems. Demonstrates a good understanding of the majority of the topics relating to thermal physics, electromagnetism, optics, atomic and nuclear physics. Has a high level of skills in the physics laboratory.

70%-100%: All the previous to an excellent level. Demonstrates an ability to put a numerical solution into a context and assess whether such solutions are meaningful. Has a very high level of efficiency in the physics laboratory. Can critically assess experiments undertaken.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Tutorial	8	
Practical	16	
Independent Learning	75	

Essential Material

- Giancoli, D. *Physics: Principles with Applications, Global Edition*. 7th ed. New Jersey: Pearson, 2015.

Supplementary Material

- Cutnell, J. and K. Johnson. *Physics*. 9th ed.. USA: Wiley, 2012.
- Halliday, D., R. Resnick and J. Walker. *Fundamentals of Physics Extended*. 10th ed.. USA: Wiley, 2013.
- Knight, R., B. Jones and S. Field. *College Physics: A Strategic Approach Technology Update*. USA: Pearson, 2015.
- Walker, J. *Physics Technology Update*. 4th ed.. New Jersey: Pearson, 2014.

Requested Resources

- SCIENCE LAB: Physics